

Estimated Time of Arrival in Autonomous Vehicles Using Gradient Boosting: Real-life case study in public transportation

Evangelos Antypas
*Information Technologies Institute
Center for Research and
Technology - Hellas
Thessaloniki, Greece
eantypas@iti.gr*

Georgios Spanos
*Information Technologies Institute
Center for Research and
Technology - Hellas
Thessaloniki, Greece
gspanos@iti.gr*

Antonios Lalas
*Information Technologies Institute
Center for Research and
Technology - Hellas
Thessaloniki, Greece
lalas@iti.gr*

Konstantinos Votis
*Information Technologies Institute
Center for Research and
Technology - Hellas
Thessaloniki, Greece
kvotis@iti.gr*

Dimitrios Tzovaras
*Information Technologies Institute
Center for Research and
Technology - Hellas
Thessaloniki, Greece
dimitrios.tzovaras@iti.gr*

Abstract—Autonomous Vehicles (AVs) are expected to revolutionise the methods of transportation. Research and innovation in this field is making huge leaps in the last few years, whether it considers vehicles used for private or public transport. Predicting the Estimated Time of Arrival (ETA) is a very important attribute associated with Public Transport (PT). Especially with the rise of AVs' adoption, PT is expected to follow this trend. Therefore, ETA prediction is deemed to be a service that interests the majority of PT stakeholders. PT is a field that automation benefits both stakeholders and commuters, and this research aims to provide a benchmark considering AVs in PT. Within this work, Gradient Boosting (GB) techniques for ETA prediction were employed, namely eXtreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost) and Light Gradient Boosting Machines (LightGBM). This study proposes competitive ETA prediction methods in Autonomous Buses, while the results of this research are very encouraging and aim to contribute to the overall investigations in the field of autonomous and automated PT.

Index Terms—Estimated Time of Arrival, prediction, XGBoost, CatBoost, LightGBM, regression, CCAM.

I. INTRODUCTION

Autonomous Vehicles (AVs) are expected to significantly increase the overall efficiency of Public Transport (PT), especially if they are equipped with digital maps, satellite navigation systems (such as GPS) and communication technologies [1]. However, user acceptance in the Cooperative, Connected and Automated Mobility (CCAM) is an open issue, as there is no conclusive work in this field. While both [2] and [3] dive into this issue, the results are not conclusive. Citizens of Mainz, Germany seem to be open minded about automating the public transport when testing the autonomous minibus EMMA [2]. On the other hand, Dutch population do not seem

to have embraced AVs as a part of their last-mile multimodal trip planning [3], when they chose second-class train trips as part of egress modes. The main concerns in both [2] and [3] seem to be the insecurity considering the driver absence and the safety issues that might arise. Meanwhile, Dutch citizens that prefer first-class train seats seem to have an average preference for using AVs as a last mile means of transportation [3]. Acceptance of AVs in PT is still an open wager, as the adoption of such transport modes have multiple advantages in a number of fields. Assuming that AVs are mainly electric-powered, the environmental impact within a city is optimistic [4]. Simultaneously, in [5], A. Tirachini and C. Antoniou have meticulously detailed the economics of PT, concluding that the cost savings due to automation exceeds the capital cost of AVs, benefiting both stakeholders and commuters.

PT plays an important role in all contemporary European cities; therefore, the precision of means of transportation is of paramount importance [6]. Travel and arrival time estimation is a precision issue that creates trust in PT, while also economically availing cities and AV providers. Estimated Time of Arrival (ETA) can be explained as the service of calculating the time needed to reach from point A to point B (e.g from one bus stop to the next). It manifests its importance for the commuters to be able to reach their destination accurately in terms of time or calculating the exact time they board the vehicle, while the vehicle providers ensure the reliability in their services and the overall cost optimisation [7].

The purpose of the present study is to enhance the CCAM paradigm by dealing with the prominent issue of predicting the ETA of AVs in PT. More specifically, this research focuses mainly on data acquired by autonomous buses with fixed

itineraries, in order to predict the time of arrival to a specific landmark in the set itinerary. Therefore, the target of this paper is to analyse the arrival time prediction problem as a regression problem and to utilise corresponding machine learning techniques to reach a conclusion. Gradient Boosting (GB) prediction methods are investigated in the present research and these are: eXtreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost) and Light Gradient Boosting Machine (LightGBM). Finally, a comparison is performed among them, in order to decide which method yields better results with a specific landmark distance for Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) metrics. This work contributes to enhanced ETA prediction for PT, in the field of Autonomous Buses, allowing optimum utilisation of a variety of competitive GB techniques.

The structure of the paper is as follows. Section II discusses the related work in estimating the arrival time with gradient boosting techniques. Section III introduces our methodology, namely the data provision and processing, along with the gradient boosting models used to tackle this issue. Section IV presents the results of our methods, as emerged from simulation methods. Section V concludes the overall work, while giving future work and research directions.

II. RELATED WORK

Prediction of bus arrival time is a topic that has been researched the last decades. While researchers have deployed a number of various approaches to tackle this problem, the emergence of automated buses caused the complexity of the previous analyses to be surged. The main traditional approaches consist of Artificial Neural Networks (NNs), Support Vector Machines (SVMs), K-Nearest Neighbors (KNNs) and Long Short Term Memory (LSTMs) networks. With the insurgence of GB Techniques, such as XGBoost, CatBoost and LightGBM, the focus has shifted from the aforementioned methods to contemporary ones. Whether the focus of each problem is mass transportation by air (airlines and shuttles) or by land (trains, buses), or even individual transportation with cars or taxis, the problem of calculating the ETA remains the same. Hence, a correct prediction of ETA is of paramount importance regardless of the means of transportation.

GB methods have been utilised in a number of studies. In [8], A. Taparia and M. Brandy use GPS data collected from a Brazilian bus route in order to compare Linear Regression, NNs, LSTM and XGBoost, using RMSE, MAE and MAPE as metrics. Moreover, in [9] and [10], the authors use data from flight transportation to compare LightGBM, XGBoost and traditional GB Trees [9], and CatBoost with Multilayer Perceptron and Bagging Regression Methods [10], respectively, with XGBoost and CatBoost outperforming the traditional methods. Similarly, in [11] the authors create a high precision estimation method of Chinese flight arrivals using XGBoost, whereas in [12], Yang et. al. provide an analysis based on comparison of LSTM, XGBoost and LSTM-XGBoost combination, tak-

ing weather data into consideration. As previously, the GB Methods seemed to be superior compared to LSTM.

Several investigations have also been conducted in railway transportation. In [13], R. Shi and X. Xu have acquired data from Chinese railways, in order to compare XGBoost with Bayesian Optimisation methods, concluding XGBoost as the method with the highest capability in solving train delay problems. In [14] Li et. al. approach the train delay forecasting and actual arrival time problems with NN, Random Forest, XGBoost, GB Decision Trees and Statistical methods. The research concluded that for this specific prediction, XGBoost comes second best after Random Forest. Additionally, considering the Tunisian railway system, H. Laifa, R. Khcherif and H. H. B. Ghezalaa [15] used SVM, Random Forest, LightGBM, XGBoost and NN approaches, and evaluated their models with the corresponding metrics RMSE and MAE. The GB methods yielded better results, with LightGBM having the best overall performance.

Concurrently, predicting the arrival time remains a predominant challenge for other means of PT [8], i.e bus itineraries, or demand-responsive transport (DRT), whether taxi routes [16] [17] or private trips [18], with applications even in the field of logistics supply chains [19]. Considering DRT, the authors of [16] collected GPS taxi data from New York and developed prediction models based on both regression and classification. The methods used are XGBoost, compared to Support Vector Regressors and NNs, as well as to TEMP-rel models and online map services. In all accounts, XGBoost models demonstrate their accuracy capability in both tasks, using MAE, MAPE and RMSE as metrics. In [17], CatBoost and XGBoost models outperform NNs, with CatBoost yielding slightly better results with MAE as the metric, using data generated by GG taxi orders. Furthermore, the studies based on freeway private trips are equally important. Indeed, the focus of [18] is ETA prediction in the Shanghai-Hangzhou-Ningbo freeway, comparing LightGBM to GB Decision Trees and K-Nearest Neighbor algorithm. Using MAE and MAPE as evaluation metrics, LightGBM was superior to the other prediction models.

In [19], the authors utilise real-time GPS data from Austria and optimisation methods for vehicle path tracking. Six variants of GB algorithms were considered, namely GB Decision Trees, Adaboost, XGBoost, LightGBM, CatBoost and Histogram GB. Their performance was compared to Random Forest, tree-bases Bagging Regressor and Linear Regression Models. The results of this research revealed a significant precedence of the GB techniques. Using MAE and RMSE as metrics, XGBoost best calculates MAE, while either XGBoost, CatBoost or LightGBM better calculate RMSE, depending on each of five scenarios the researchers take into account.

It is made clear by the aforementioned analysis that -to the best of our knowledge- there are not (until now) research studies in the respective literature that deal with the crucial issue of ETA in autonomous vehicles. Hence, considering that the field of AVs is emerging and that there will be a burgeoning adoption of CCAM the next years, rendering the

need for respective CCAM service indispensable, we aspire to produce a novel proposal for AV's ETA as a baseline for future research.

III. METHODOLOGY

For the context of this paper, all the data were acquired by the Linköping pilot site, within the SHOW project ecosystem [20]. The AV transmits data regarding its location, speed, acceleration, the timestamp of each entry and sparsely transmits the outdoor weather. Figure 1 shows the itinerary followed by the Linköping pilot site. The data were obtained via a real-time MQTT connection [21], with frequency of one entry per second. The acquired timestamp was then converted to date, time of the day and the cumulative seconds within that day.

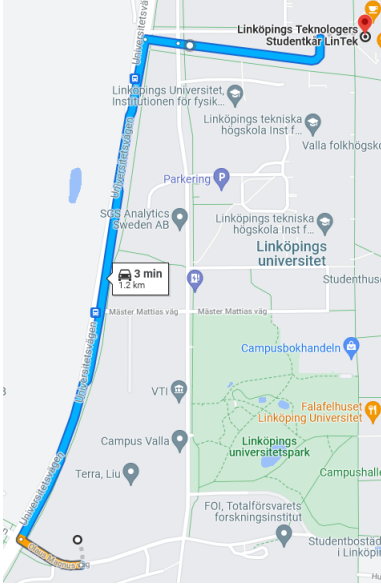


Fig. 1. The itinerary of the Linköping pilot site's Autonomous Vehicle.

This section first discusses the data gathering and processing, while it elaborates the choices conducted in this regard. Afterwards, the algorithms used in this scope are discussed, in order to solve the ETA regression problem. Finally, the metrics that evaluate the proposed methodology are presented.

A. Data Processing

The first challenge for the present study was to calculate the distance between two specific points. It is important to note that the SHOW H2020 project [20] is still in process, therefore no fixed stations were decided by the pilot site for the testing phase. Given that the AV does not have any stops within its original itinerary, the appropriate distance should be defined, for calculating the arrival time. As the proposed distance according to the literature for bus stops within urban areas should not exceed 300 meters [22], the stops were placed within 100, 200 and 300 meter distance from each other. One extra scenario was considered, with 400 meter stop distance, as this is also typical in real-life itineraries [22]. Thus, in this itinerary of 1.2 km (as seen in Figure 1), a total of twelve,

six, four and three stops were created respectively. The next step was to pinpoint the indices of the data, where the AV had traversed the desired distances. For this problem, the haversine formula was used.

$$d = 2r \arcsin \sqrt{\sin^2 \left(\frac{\phi_2 - \phi_1}{2} \right) + \cos \phi_1 \cdot \cos \phi_2 \cdot \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \quad (1)$$

In (1), d is the calculated distance, r is the earth's radius, while ϕ_1 , ϕ_2 , λ_1 and λ_2 denote the latitudes and longitudes of points 1 and 2 respectively. Algorithm 1 was developed and implemented to calculate the desired points and distances.

Algorithm 1 Haversine Calculation and index pinpoint

```

while ( $i < \text{data.length}$ ) do
   $\text{table} \leftarrow i$ 
   $\text{lon}_1 = \text{longitude}_i$ 
   $\text{lat}_1 = \text{latitude}_i$ 
  for ( $j$  in [ $i+1$ ,  $\text{data.length}$ ]) do
     $\text{lon}_2 = \text{longitude}_j$ 
     $\text{lat}_2 = \text{latitude}_j$ 
    (1)  $\text{haversine} \leftarrow \text{lat}_1, \text{lat}_2, \text{lon}_1, \text{lon}_2$ 
    if ( $\text{haversine} > \text{stop\_distance}$ ) then
       $\text{data} \leftarrow \text{haversine}$ 
      BREAK FOR LOOP
    end if
  end for
   $i = j + 1$ 
end while
return  $\text{table}$ 
return  $\text{data}$ 

```

In Algorithm 1, a table is formed that stores indices. The latitude and longitude of the first entry are obtained, while the algorithms searches for the location whose latitude and longitude reaches the desired distance. The index of the second location is stored in the table and the calculated haversine is added to the dataset. Then the search is continued, taking the following entry as the starting point and searching for the next entry whose location is orientated at the desired distance, and so forth until the end of the dataset. Therefore, we obtain a table with the indices of the desired locations, which can be isolated from the dataset, along with the calculated haversine distances.

In the next step, we keep the data with the desired indices, as acquired from the table emerged from Algorithm 1. Then the differences between the time of the isolated data are calculated (the first entry of every day is omitted, as no distance was covered in this itinerary). In case of a latency, which is essentially a sign of abnormality, two scenarios were considered. In the first scenario, if the elapsed time between two stops is more than 10 minutes, then the entry was considered as noise outlier and was deleted. The second scenario follows the same premise, but the elapsed time was set to 20 minutes. This corresponds to the data received from the AV, as no reports of incidents or abnormal activities (e.g accidents, vehicle breakdown etc.) are received, even though

the AV has that ability. Therefore, we deduced that in case of such latency, the vehicle was not active during that specific time period. Nevertheless, the next stage of implementation (within the SHOW project) is to automate this procedure and receive real-time notifications in case such abnormal events occur.

Finally, a zoning method was considered to classify days and time-zones. Each day of the week was appointed a day number (1 to 7), while the time of vehicle activity was split in three (one in the morning, 7:53:39 to 12:43:07, one in the afternoon, 12:43:07 to 17:45:37 and one where there was no shuttle operation). This classification was operated according to the accumulated data, where the earliest entry was at 7:53:39, the latest at 17:45:37 and 12:43:07 was the statistical average. This method assists with the discovery of a potential periodicity in case of a regular time-event recurrence. Such applications are very important in scenarios within cities with specific traffic patterns, as both arrival time and demand forecasting due to traffic are highly influenced by day and time-zone periodicity. Table I presents an overall overview of the variables used in this regression problem, taking a random entry from the dataset.

TABLE I
DATA OUTPUT AND INPUT FOR REGRESSION ANALYSIS

Variable	Value
longitude	15.572918
latitude	58.397013
speed (m/s)	2.232531
acceleration (m/s ²)	0.701622
temperature °C	7.0
time zone	2
day zone	4
haversine (m)	200.619
elapsed time (s)	121.0

B. Gradient Boosting Techniques

Calculating ETA can be formulated as a regression problem. For this purpose, regression methods utilising Gradient Boosting Regression Trees (GBRT) were implemented. As discussed thoroughly in Section II, XGBoost, CATBoost and LightGBM are three methods that vastly outperform, among other machine learning methods, conventional GB regression solutions. It is consequently considered reasonable for this research to be based on these solutions. A detailed comparison among these models is also provided, for a number of scenarios.

The GBRT algorithms family works as follows [23] [24]. Having a dataset $D = \{x_i, y_i\}_1^N$, GBRT aim to create approximations $\hat{F}(x)$ of the function $F(x)$. $F(x)$ matches the outcomes x to the output values y , by minimising the expected value of a loss function $L(y, F(x))$. In our research, the loss function is RMSE.

$$F_m(x) = F_{m-1}(x) + \rho_m h_m(x) \quad (2)$$

$$F_0(x) = \underset{a}{\operatorname{argmin}} \sum_{i=1}^N L(y_i, a) \quad (3)$$

$$(\rho_m, h_m(x)) = \underset{\rho, h}{\operatorname{argmin}} \sum_{i=1}^N L(y_i, F_{m-1}(x_i) + \rho h(x_i)) \quad (4)$$

GB methods create a cumulative approximation of $F(x)$ as weighted sum of (2), where ρ_m represents the weight of the m^{th} base learner model function, $h_m(x)$. The first constant F_0 is calculated by (3), while the rest of the models minimise (4). Nevertheless, each h_m can be thought of as a greedy step in a gradient descent optimisation for $F(x)$ where each h_m is trained on another dataset $D' = \{x_i, r_{mi}\}_{i=1}^N$. The pseudo-residuals r_{mi} are calculated by (5).

$$r_{mi} = \left[\frac{\partial L(y_i, F(x))}{\partial F(x)} \right]_{F(x)=F_{m-1}(x)} \quad (5)$$

a) *eXtreme Gradient Boosting*: Tianqi Chen and Carlos Guestrin [25] first proposed the XGBoost method, as a scalable end-to-end tree boosting system, inspired by Sparsity-aware Split Finding Algorithms. The goal of XGBoost is to minimise a loss function (6), in order to build an additive expansion of the objective function [23] [24].

$$L_{XGBoost} = \sum_{i=1}^N L(y_i, F(x_i)) + \sum_{m=1}^M \Omega(h_m) \quad (6)$$

$$\Omega(h) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (7)$$

In (7), T denotes the number of leaves and w are the output scores of leaves, while γ governs the loss reduction needed in order to split internal nodes (the higher the values of γ , the simpler the trees) [23] [25].

b) *Categorical Boosting*: CatBoost addresses the problem of prediction shift in GB [26], which occurs when GB algorithms use the same instances for estimating both the gradients and the models that minimise the aforementioned gradients, called target leakage [23] [26]. Inspired by GBRT, having a dataset $D = \{x_i, y_i\}_1^N$, the aim is to train a function $F : \mathbb{R}^m \rightarrow \mathbb{R}$, in order to minimise the expected loss $L_{CatBoost} = \mathbb{E}L(y, F(x))$. $L(\cdot, \cdot)$ denotes a smooth loss function, while (x, y) is a test example sampled from the training dataset D [26].

A sequence of approximations $F^t : \mathbb{R}^m \rightarrow \mathbb{R}$, $t \in \mathbb{N}_0$, is constructed by GB techniques in a greedy manner [26] [24]. F^t is obtained from F^{t-1} in a cumulative fashion, such as $F^t = F^{t-1} + \alpha g^t$, using α as a step size and a base predictor function $g^t : \mathbb{R}^n \rightarrow \mathbb{R}$. The goal of the base predictor is to minimise the expected loss and is chosen from a superset of functions G [26]:

$$g^t = \underset{g \in G}{\operatorname{argmin}} L(F^{t-1} + g) = \underset{g \in G}{\operatorname{argmin}} \mathbb{E}L(y, F^{t-1}(x) + g(x)) \quad (8)$$

A common method of approaching this minimisation problem is either to use the Newton method with a second-order approximation of $L(F^{t-1} + g)$ at F^t , or by taking a negative gradient step, since this is the norm for regression trees [26].

c) *Light Gradient Boosting Machines*: Ke et al. [27] proposed LightGBM as a free and open source extensive library to implement GB methods with extensive variants. This solution is based on decision trees algorithms and targets at creating a computationally efficient algorithm. LightGBM method consist of two GB Decision Trees (GBDT) features, Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). EFB groups data with mutually exclusive features, in order to reduce the dimensionality of the problem and therefore improve efficiency [27] [23]. GOSS takes advantage of the fact that there is no native weight for data instance in GBDT. Instances with small gradients produce smaller training errors and the ones with larger gradients are undertrained. This approach makes undertrained instances contribute more to the overall information gain, by keeping the ones with larger gradients while randomly dropping others with small gradients [27]. Such treatment leads to more accurate gain estimation than random sampling. GOSS method is the one used in the scope of this research.

C. Evaluation Metrics

The evaluation metrics used in this scope are as follows. Being consistent with the literature and related work, as described in Section II, the corresponding metrics are RMSE, MAE and MAPE. These metrics were used for the problem investigated in this work, to measure the performance of the methods described in Section III-B.

- *Root Mean Square Error*. RMSE is an estimator that measures the root of the average squared difference of the estimated values from the actual values, i.e the square root of the average of the squares of errors.
- *Mean Absolute Error*. MAE measures the absolute difference of the estimated values from the actual values.
- *Mean Absolute Percentage Error*. MAPE measures the difference of the estimated values from the actual values, while expressing the accuracy as a ratio.

For the scope of this research, the python3 programming language was used and specifically, the corresponding packets for the aforementioned GBRT were xgboost [28] v.1.5.1, catboost [29] v.1.0.3 and lightgbm [30] v.3.3.2.

IV. RESULTS

In this section, the results of this research are presented. As mentioned in Section III-B, three main GB methods were used to approach the problem of arrival time estimation concerning AVs in PT. The results are presented in an aggregate manner, as a comparison among XGBoost, CatBoost and LightGBM for a number of scenarios that will be analysed in the following paragraphs.

For the aforementioned Machine Learning techniques, the following evaluation procedure was considered. As mentioned in Section III-A, the desired travel distances were set to 100,

200, 300 and 400 meters, while latencies exceeding 10 minutes (scenario 1) or 20 minutes (scenario 2) are omitted. Moreover, specific periodic activity was taken into consideration, regarding days of the week and time-zones of the day. The data were split in an efficient manner, as data acquired consists of 13 days from the beginning of December 2021. Therefore, 11 days for the training set and 2 days for the test set were used, constructing an 85% - 15% split for the purposes of this paper. The desired attributes that contribute in the training phase, are the vehicle's latitude and longitude, its speed and acceleration values, as well as the outdoor temperature. The desired prediction (regression) variable is the time elapsed between covering the desired distances, measured in seconds. For all the aforementioned scenarios, the performance of XGBoost, CatBoost and LightGBM in terms of the evaluation metrics was particularly high. However, since the purpose of the study is the enhancement of CCAM paradigm by estimating the arrival time of AVs and not a comparative study between different machine learning algorithms, such a comparison does not contribute to the overall goals. Hence, the presentation of the results tries to highlight the overall high performance of the methodology independently of the algorithm used.

TABLE II
GRADIENT BOOSTING ALGORITHMS COMPARISON FOR 100 METERS

100 meters	Evaluation Metrics		
10 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	48.12	22.21	32.01
LightGBM	51.12	28.08	48.50
CatBoost	49.32	22.06	31.90
20 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	53.70	26.61	45.46
LightGBM	74.72	45.03	95.50
CatBoost	48.88	27.28	50.13

TABLE III
GRADIENT BOOSTING ALGORITHMS COMPARISON FOR 200 METERS

200 meters	Evaluation Metrics		
10 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	51.73	31.35	29.33
LightGBM	61.88	39.48	33.22
CatBoost	57.02	33.55	29.78
20 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	79.06	39.01	41.52
LightGBM	90.41	63.81	61.21
CatBoost	69.30	44.24	44.31

Tables II, III, IV and V display the results according to the evaluation metrics for all eight scenarios previously described (with regard to latency and stop distance) and for the three GB methods. By inspecting these tables, it is easily deductable that MAE is less than a minute and the more strict RMSE is below 100 seconds in most of the considered scenarios, while the MAPE is typically less than 50% (achieving in the most common scenario for the case of AVs of 200 meters and 10 minutes latency, percentage error around 30% and mean

TABLE IV
GRADIENT BOOSTING ALGORITHMS COMPARISON FOR 300 METERS

300 meters	Evaluation Metrics		
10 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	78.25	48.34	31.48
LightGBM	80.26	51.92	24.68
CatBoost	94.84	60.81	35.77
20 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	94.26	57.79	38.10
LightGBM	96.45	69.32	46.17
CatBoost	90.72	57.82	38.18

TABLE V
GRADIENT BOOSTING ALGORITHMS COMPARISON FOR 400 METERS

400 meters	Evaluation Metrics		
10 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	78.43	50.83	28.67
LightGBM	85.11	57.85	35.23
CatBoost	74.35	51.01	28.85
20 minutes	RMSE (seconds)	MAE (seconds)	MAPE (%)
XGBoost	104.93	69.01	35.66
LightGBM	141.59	106.18	72.47
CatBoost	133.36	92.68	64.54

error approximately half a minute). The delay of half a minute to a minute for each stop and the low percentage errors are considered as encouraging result.

Moreover, using larger distances between stops tends to yield better results, as does penalising latency larger than 10 minutes, compared to the 20 minute scenario. The scenario of 400 meters / 20 minutes summons the overall worst results, disrupting the trend of the reduction of MAPE with the distance increase. However, even in this case, MAPE gets the maximum value of 72%. Therefore, calculating the arrival time between reasonably longer distances, as well as being stricter on how much time it is considered that the vehicle is active, could reveal a methodology for a commercial ETA prediction with real-life applications.

V. CONCLUSION

This work engages with the impact of AVs in PT. While the field of AVs is emerging, with a number of studies focusing on privately owned cars, the impact of such advancement of PT is still under-researched. A number of researches and services are ought to be carried out, in order to fulfill the need of a number of passengers that either do not own a private vehicle or do not intend to use it for commuting.

In this scope, estimating the arrival time of vehicles is of paramount importance, as punctual arrival to the destination is the prevailing goal of each itinerary. To tackle this issue, which is approached as a regression problem, a methodology based on three different methods of GB Algorithms was implemented. The data were acquired by a real life autonomous shuttle and an algorithm was proposed to calculate the distance covered in a real-time manner, with frequency of one entry per second. The GB techniques used in this scope were XGBoost,

CatBoost and LightGBM, utilising the Gradient-based One-Side Sampling approach. It is undeniable that the present study achieved its purpose since all methods produce encouraging results as reflected in Section IV, resulting in low percentage and mean errors and for the scenarios considered. Having acquired such encouraging results, the contribution of this research, along with the potential of enhanced ETA for PT is justified. Therefore, the yielded results of this paper could be a basis for a radical shift towards increasing passenger adherence and accelerating Autonomous Mobility in PT. This work aspires to be considered as a benchmark approach for AVs in PT, with regard to time prediction approaches.

However, further work should be carried out in the future, for a number of different scenarios. Even more extensive parameter tuning for each scenario could provide even better results. In more details, future work could also include having real-time notifications of arrival time, when fixed bus stops are introduced in an itinerary. The most prominent point for future work is the validation of the proposed methodology performance by expanding this research in a number of cities for different itineraries.

Furthermore, better results could arise combining the proposed ETA with a system that will notify in the case of an unexpected event, therefore excluding these entries from the calculation. Moreover, even though in this scope there exists low latency due to traffic, in a real life scenario where urban transportation is considered, different time zones have different traffic and therefore affect arrival time, while bus acceleration, deceleration and speed, when bus stops are taken into account, are also important factors. Interaction with other objects and mixed traffic situations within the itinerary could also pose an obstacle to overcome. Thus, ETA prediction should take such examples into consideration. Finally, it is noted throughout this document that the specific algorithms used in this scope are really efficient, however future research could include simpler and faster algorithms as benchmark, or even more combined and sophisticated Machine Learning and Deep Learning algorithms.

ACKNOWLEDGMENT

This work was funded by the European's Union Horizon 2020 Research and Innovation Program through SHared automation Operating models for Worldwide adoption (SHOW) project under Grant Agreement No. 875530.

REFERENCES

- [1] P. Smietanka, K. Szczypiorski, F. Viti, and S. Marcin, "Distributed automated vehicle location (avl) system based on connected vehicle technology," in *2015 IEEE 18th International Conference on Intelligent Transportation Systems*. IEEE, 2015, pp. 1946–1951.
- [2] C. Bernhard, D. Oberfeld, C. Hoffmann, D. Weismüller, and H. Hecht, "User acceptance of automated public transport: Valence of an autonomous minibus experience," *Transportation research part F: traffic psychology and behaviour*, vol. 70, pp. 109–123, 2020.
- [3] M. D. Yap, G. Correia, and B. Van Arem, "Preferences of travellers for using automated vehicles as last mile public transport of multimodal train trips," *Transportation research part a: policy and practice*, vol. 94, pp. 1–16, 2016.

- [4] E. F. Choma and C. M. L. Ugaya, "Environmental impact assessment of increasing electric vehicles in the brazilian fleet," *Journal of Cleaner Production*, vol. 152, pp. 497–507, 2017.
- [5] A. Tirachini and C. Antoniou, "The economics of automated public transport: Effects on operator cost, travel time, fare and subsidy," *Economics of Transportation*, vol. 21, p. 100151, 2020.
- [6] M. Luethi, U. Weidmann, and A. Nash, "Passenger arrival rates at public transport stations," in *TRB 86th Annual Meeting Compendium of Papers*. Transportation Research Board, 2007, pp. 07–0635.
- [7] M. U. Farooq, A. Shakoor, and A. B. Siddique, "Gps based public transport arrival time prediction," in *2017 International Conference on Frontiers of Information Technology (FIT)*. IEEE, 2017, pp. 76–81.
- [8] A. Tapania and M. Brady, "Bus journey and arrival time prediction based on archived avl/gps data using machine learning," in *2021 7th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)*. IEEE, 2021, pp. 1–6.
- [9] J. Tao, H. Man, and L. Yanling, "Flight delay prediction based on lightgbm," in *2021 IEEE 3rd International Conference on Civil Aviation Safety and Information Technology (ICCASIT)*. IEEE, 2021, pp. 1248–1251.
- [10] X. Dou, "Flight arrival delay prediction and analysis using ensemble learning," in *2020 IEEE 4th Information Technology, Networking, Electronic and Automation Control Conference (ITNEC)*, vol. 1. IEEE, 2020, pp. 836–840.
- [11] G. Wang, K. Liu, H. Chen, Y. Wang, and Q. Zhao, "A high-precision method of flight arrival time estimation based on xgboost," in *2020 IEEE 2nd International Conference on Civil Aviation Safety and Information Technology (ICCASIT)*. IEEE, 2020, pp. 883–888.
- [12] Z. Yang, Y. Wang, J. Li, L. Liu, J. Ma, and Y. Zhong, "Airport arrival flow prediction considering meteorological factors based on deep-learning methods," *Complexity*, vol. 2020, 2020.
- [13] R. Shi and X. Xu, "A train arrival delay prediction model using xgboost and bayesian optimization," in *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2020, pp. 1–6.
- [14] Z. Li, C. Wen, R. Hu, C. Xu, P. Huang, and X. Jiang, "Near-term train delay prediction in the dutch railways network," *International Journal of Rail Transportation*, vol. 9, no. 6, pp. 520–539, 2021.
- [15] H. Laifa, H. H. B. Ghezalaa *et al.*, "Train delay prediction in tunisian railway through lightgbm model," *Procedia Computer Science*, vol. 192, pp. 981–990, 2021.
- [16] K. D. Kankanamge, Y. R. Witharanage, C. S. Withanage, M. Hansini, D. Lakmal, and U. Thayasivam, "Taxi trip travel time prediction with isolated xgboost regression," in *2019 Moratuwa Engineering Research Conference (MERCon)*. IEEE, 2019, pp. 54–59.
- [17] H. T. Sergoyan, "Estimating time of driver arrival with gradient boosting algorithms and deep neural networks," *Mathematical Problems of Computer Science*, vol. 53, pp. 29–38, 2020.
- [18] F. Wang, H. Cheng, H. Dai, and H. Han, "Freeway short-term travel time prediction based on lightgbm algorithm," in *IOP Conference Series: Earth and Environmental Science*, vol. 638, no. 1. IOP Publishing, 2021, p. 012029.
- [19] J. Khiari and C. Olaverri-Monreal, "Boosting algorithms for delivery time prediction in transportation logistics," in *2020 International Conference on Data Mining Workshops (ICDMW)*. IEEE, 2020, pp. 251–258.
- [20] Shared automation operating models for worldwide adoption. Accessed: 2022-05-31. [Online]. Available: <https://show-project.eu/>
- [21] A. Sersemis, A. Papadopoulos, G. Spanos, A. Lalas, K. Votis, and D. Tzovaras, "A novel cybersecurity architecture for iov communication," in *25th Pan-Hellenic Conference on Informatics*, 2021, pp. 357–361.
- [22] A. Tirachini, "The economics and engineering of bus stops: Spacing, design and congestion," *Transportation research part A: policy and practice*, vol. 59, pp. 37–57, 2014.
- [23] C. Bentéjac, A. Csörgő, and G. Martínez-Muñoz, "A comparative analysis of gradient boosting algorithms," *Artificial Intelligence Review*, vol. 54, no. 3, pp. 1937–1967, 2021.
- [24] A. A. Ibrahim, R. L. Ridwan, M. Muhamme *et al.*, "Comparison of the catboost classifier with other machine learning methods," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 11, pp. 738–748, 2020.
- [25] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 2016, pp. 785–794.
- [26] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "Catboost: unbiased boosting with categorical features," *Advances in neural information processing systems*, vol. 31, 2018.
- [27] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "Lightgbm: A highly efficient gradient boosting decision tree," *Advances in neural information processing systems*, vol. 30, 2017.
- [28] extreme gradient boosting. Accessed: 2022-05-31. [Online]. Available: https://xgboost.readthedocs.io/en/stable/python/python_intro.html
- [29] Categorical boosting. Accessed: 2022-05-31. [Online]. Available: <https://catboost.ai/en/docs/concepts/python-quickstart>
- [30] Light gradient boosting machines. Accessed: 2022-05-31. [Online]. Available: <https://lightgbm.readthedocs.io/en/latest/Python-Intro.html>